



MINING DEALER BEST PRACTICE

Demand Planning with AMT

**Maintenance and
Repair**

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1802-1.07-1214**

1.0 Introduction

The purpose of this best practice is to document and communicate how to use the 3rd party software AMT (Asset Management Tool) for mining parts demand planning. AMT is a server or cloud-based program (both options available depending on the need of the user) that manages the site assets for a customer. The system has been adopted by various dealers around the world to manage mining assets, including those under a MARC (Maintenance & Repair Contract), and provides total dealer territory component visibility. This tool was developed by iSolutions, which was then acquired by RPM (Runge Pincock Minarco) in 2016.

2.0 Best Practice Description

OVERVIEW

The demand planning best practice document provides AMT system user understanding to support each customer's asset needs and have clearer visibility to those product support opportunities within a dealer's territory. Adoption & utilization of AMT systems by dealers provides a central planning hub that can be used and shared between internal dealer departments.

This internal process should aim to capture field validated data that is utilized across the dealership to make sound business decisions. The process owner must ensure departmental engagement and a periodic & cyclical process should be embedded into the business to ensure the data housed in AMT is current and maintained.

SYSTEM

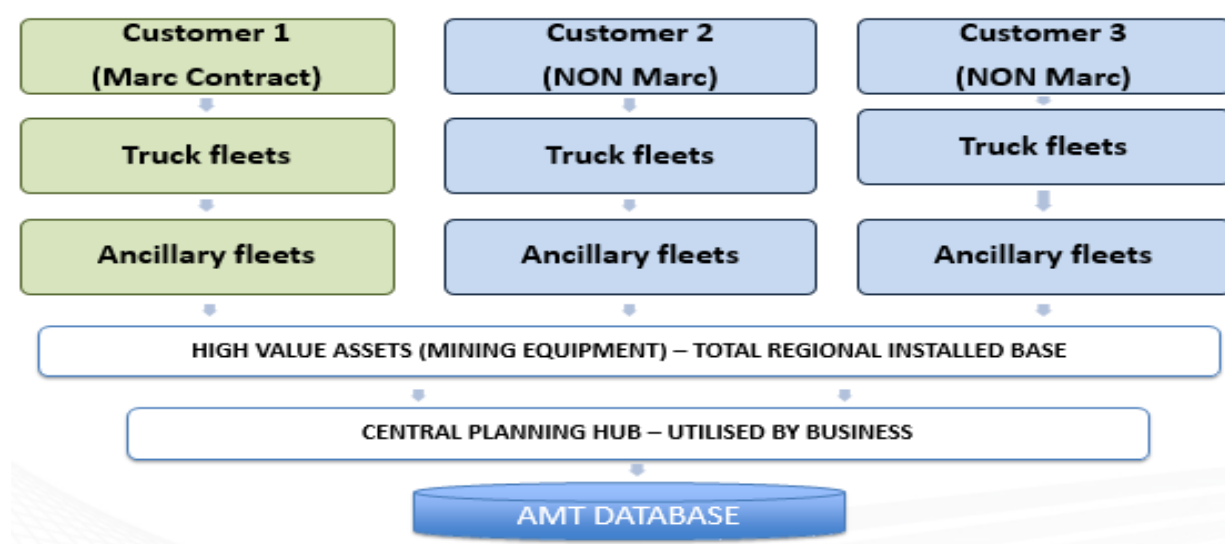
In a dealer's territory where a large installed base of production machines exists, the benefit of a tool such as AMT becomes more apparent. Regional population in scope (decided by the dealer and based on the points below) for demand planning should be loaded into the system. In the past only MARC machines were loaded & managed, but that mindset has since changed for several dealers to now load / maintain all mining machines in the territory (i.e. both MARC and non-MARC).

Assets generally loaded and maintained in AMT would include:

- Customer mining machines
- Machines under MARC and non-MARC
- Assets primarily used for production or to support production

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Below is a graphic of the type of data that would be brought into AMT for demand planning purposes:



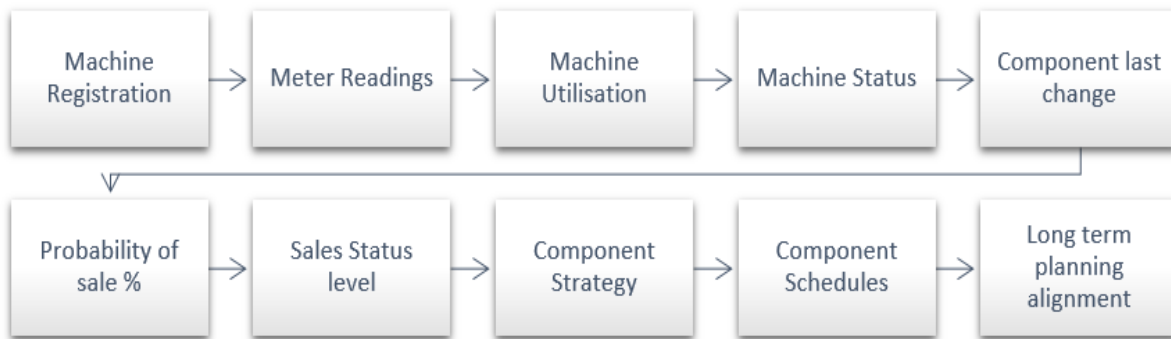
VALIDATION

After the assets are loaded and set up in the system, periodic field validation and AMT system maintenance is required to ensure the data is updated. Although the frequency is determined by the dealer, it is recommended to perform this maintenance / validation at least once per month. The data requiring an update would include:

- Meter reading (SMU – Service Meter Units)
- Machine utilization (trend)
- Machine status (i.e. Operating, parked, scrapped, sold)
- Major component last change-out details (updated)
- Customer sales status and / or Customer probability of sale for upcoming opportunity events (%)
- Customer's component maintenance strategy (i.e. Rebuild, Reman, etc.) (Part classification)
- Primary part number of the major component for the specific machine serial number (PPN – Primary Part Number)
- Component schedule understanding / alignment with customer (based on latest site plan and condition monitoring)
- System adjustments in long term planning, aligned with customers plans

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PROCESS FLOW



COMPONENT STRATEGY

Once the customer / dealer component strategy has been validated & maintained in AMT, a forward-looking dealer territory forecast can be discussed and shared across dealer departments. Three elements of analysis are required as below:

- 1) Determine whether PEX (Dealer Parts Exchange) stock level is adequate, considering CRC (Component Rebuild Centre) component rebuild turnaround time
- 2) Determine the piece parts that will be required for the rebuilds using local dealer builder files & standard jobs
- 3) Understand customer requirements & CRC maximum capacity to determine if peak shaving is needed by utilizing Reman major components to support turnaround times & machine availability requirements at site
- 4) Order the necessary parts and/or components and plan the necessary labor & tooling requirements to support the rebuild / replacement events
- 5) Collaborate with the dealer's product support commercial and Global Aftermarket Solutions teams to improve the probability of sale %'s for those that have currently been classified as less than 50%

PCR / PLESN

On a quarterly basis, Caterpillar requests a validated, forward-looking PCR / PLESN (Planned Component Replacement / Plan for Every Serial Number) forecast for a two-year period from dealers. Here, validation means the dealer has spoken with the customer and verified the most current site rebuild / replacement plans for the serial numbers in question, rather than just using a theoretical calculation to estimate the rebuild / replacement plans. These dealer territorial volumes are aggregated into a global perspective to better understand where any inventory adjustments are necessary. From an inventory standpoint, the Cat Inventory Management (IM) team compares the on-hand and on-order (with our supply base) levels by part number to the forecasted demand being reflected in PCR & PLESN. IM then makes the decision to adjust schedules on the supply base as needed to meet those projected demand levels. After inventory adjustment decisions have been made by IM, the data is categorized by value (US\$ Dealer Net) by Cat distribution center and by product line, and provided back to the dealers on a quarterly basis for those that participated in the particular quarter's cycle. It should be mentioned that all mining dealers are strongly encouraged to participate in each

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quarter's cycle to ensure the supply base is understanding & receiving accurate and updated signals from the field regarding what parts & quantities will be needed and when.

Also included in the process should be the logical outcome where dealers are placing planned orders (CPRO – Committed Planned Replacement Order; or FDO – Future-Dated Order) that align with the PCR / PLESN signals received from the site. By placing planned orders that align with these signals, dealers can improve the probability their order will be filled when needed, given planned orders have a higher priority in the system (i.e. CPRO order type is the highest priority order other than critical life and/or war-type situations).

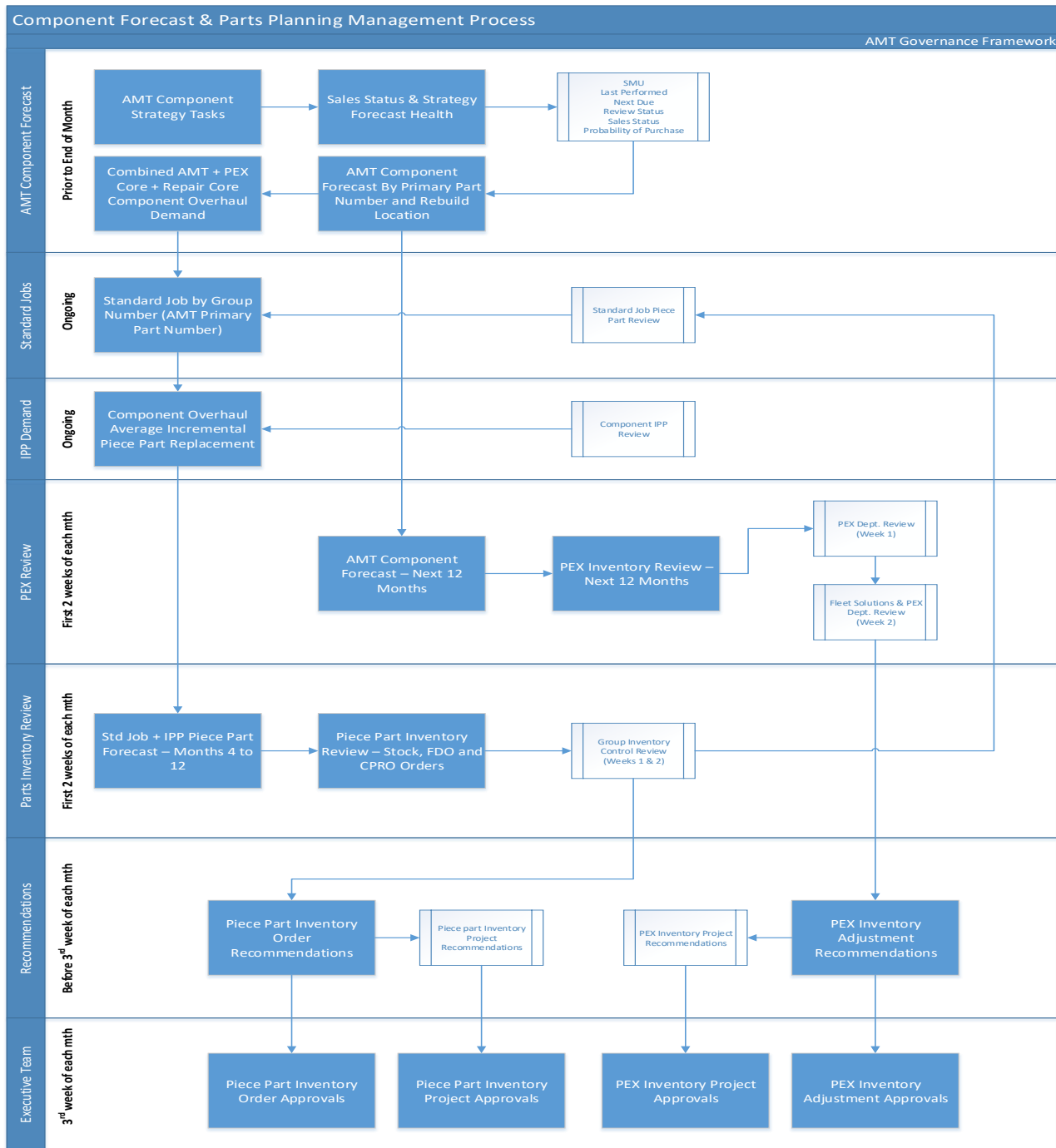
Additionally, the PCR / PLESN information provided by dealers is utilized by other internal Cat divisions such as product groups (i.e. Lifecycle analysis, projected opportunities, repair option gaps & potential pricing reviews) and regional teams (i.e. Address lower probability of sale opportunities to further work these and close to our favor).

EXAMPLE OF DEALER FORECASTING PROCESS (Hastings Deering AUS / PNG)

The process outlined on the following page showcases the involvement of a number of key dealer functions in order to both produce the component and part forecast as well as incorporate this into an overall dealer Sales & Operational Planning (S&OP) process. The outcome of this process is the placement of orders on Cat to meet forecasted customer demand for those parts and components the dealer currently does not have in inventory.

An additional acronym used in the process map below that has not yet been mentioned is IPP. This stands for Incremental Piece Parts, and involves Hastings Deering's process / analysis of identifying the average replacement rate for piece parts used in the component rebuilds.

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3.0 Implementation Steps

Implementing this Best Practice will require the commitment of the following key stakeholders to ensure success.

- 1) Dealer senior level management responsible for:
 - a) Committing to the vision of demand planning
 - b) Cascading this vision to lower levels within the dealership to ensure all key departments are on board and aligned
 - c) Empowering the individuals responsible for executing demand planning with the necessary resources to be successful
- 2) Customer facing personnel responsible for (could be PSSR's (Parts & Service Sales Representatives), account representatives, site planners, or other site personnel):
 - a) Collecting customer maintenance planning information and repair strategies to be pursued
 - b) Updating component replacement schedules, machine meter readings and actual component replacement data (serial number, major component part number, date when service was performed, hours on component when service was performed)
 - c) Maintaining the estimated probability of sale %'s and Sales Status for future component replacements
- 3) Service Operations personnel responsible for:
 - a) Ongoing Standard Job development
 - b) IPP parts identification through average work order analysis routines to ensure these are regularly updated as component lives are extended and the associated replacement %'s of the required piece parts can fluctuate
- 4) AMT administration personnel responsible for:
 - a) Performing regular analysis of AMT Forecast
 - b) Bringing together the component forecast and the piece part requirements
 - c) Sharing this data across the dealership to the impacted departments to ensure all departments are leveraging the same data / source and planning accordingly
- 5) Parts Department personnel responsible for:
 - a) Analyzing the component replacement requirements against PEX inventory levels to determine Cat Reman peak shaving needs, and placing CPRO's accordingly and as needed
 - b) Analyzing and placing CPRO orders for required piece parts to meet the expected CRC rebuild demand
- 6) CRC and/or Service Department personnel responsible for:
 - a) Analyzing the expected rebuild signals from AMT to ensure sufficient capacity exists to address the demand
 - b) Planning labor and tooling to align with the expected rebuild demand
- 7) Product Support Sales Department personnel responsible for:

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- a) Analyzing the AMT data where the probability of sale %'s for certain opportunities are lower than expected
- b) Collaborating with the necessary partners such as Global Aftermarket Solutions to address the lower probability % and ensure steps are being taken to close the opportunity in our favor (i.e. assurance, bundling, CSA, pricing, etc.).

Successfully bringing this process together requires an overall S&OP management process to manage all steps as the end-to-end process cannot be successfully performed in isolation by a single functional department.

4.0 Benefits

The benefits of a robust demand planning process are not only for Caterpillar and the dealer, but extend to the customer as well, providing a win/win/win scenario. The graphic below highlights the high-level benefits for each party, though these lists are not exhaustive.



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5.0 Resources Required

The resources required to establish a robust demand planning process depends on the process, people, and system to be leveraged. The process has been laid out in this best practice, and the people mentioned above need to have the necessary time to carry out these functions. From a system standpoint, AMT has been mentioned here as a tool that has worked well for several large mining dealers to support the demand planning process. If AMT is considered for a dealer, a decision needs to be made whether to go with the more-expensive server option, or lesser-expensive (and fewer capabilities) Cloud-based option. For further consideration and discussion, the dealer can contact the regional Global Aftermarket Solutions Demand Planner.

6.0 Supporting Attachments / References

Demand planning best practice supporting screen shot document_AMT.pptx
<https://dealer.cat.com/en/ps/parts/partsgrams.html> (parts grams regarding CPRO and FDO order types)

7.0 Related Best Practices

F-Series Truck Rebuild Parts Availability - Best Practice Document

8.0 Acknowledgements

Special thanks to Hastings Deering (Australia) for sharing valuable information to support this best practice.

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